DESIGNATIONS AND ABBREVIATIONS

AI	Artificial Intelligence
ALO	Attention–Lesion Overlap (explainability metric)
APTOS	Asia Pacific Tele-Ophthalmology Society (2019 Blindness Detection dataset)
AUC	Area Under the Curve
BYOL	Bootstrap Your Own Latent (self-supervised learning method)
CAM	Class Activation Mapping
CFC-n	Claim Formulation Constraint n — a rule governing the form a claim may take; the CFC-2 series enumerates claim types this dissertation forbids itself
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CNR	Contrast-to-Noise Ratio
DDR	Diabetic Retinopathy grading dataset (DDR)
DGL-n	Deployment and Generalization Limitation n — a bound on how far a result may be carried beyond the conditions under which it was obtained
DINO	self-Distillation with No labels (self-supervised learning method)
DR	Diabetic Retinopathy
ECE	Expected Calibration Error
EH-n	Evidence Hierarchy rule n — the ranking of metrics by evidentiary weight, and the criteria a result must meet before it counts as decisive
EHR	Electronic Health Record
EMD	Earth Mover's Distance
EyePACS	Eye Picture Archive Communication System (diabetic-retinopathy dataset)
FOV	Field of View
GDPR	General Data Protection Regulation
Grad-CAM	Gradient-weighted Class Activation Mapping
HIPAA	Health Insurance Portability and Accountability Act
IDRiD	Indian Diabetic Retinopathy Image Dataset
IoU	Intersection over Union
KL	Kullback–Leibler divergence
LAB	CIELAB colour space
MLP	Multilayer Perceptron
MoCo	Momentum Contrast (self-supervised learning method)
NC-n	Non-Claim n — a statement this dissertation explicitly does not make, recorded so that its results are not read as making it
NPV	Negative Predictive Value
OD	Optic Disc
OD-n	Operational Definition n — the definition by which a term of this

	dissertation is measured; distinct from OD above, which is never written with a number
ODIR-5K	Ocular Disease Intelligent Recognition dataset (5,000 patients)
PACS	Picture Archiving and Communication System
PC-n	Primary Claim n — a numbered claim of the dissertation's argument, each carrying its evidence and its assessed strength
PPV	Positive Predictive Value
ReLU	Rectified Linear Unit
RFMiD	Retinal Fundus Multi-disease Image Dataset
RIQA	Retinal Image Quality Assessment
ROC	Receiver Operating Characteristic
SB-n	Scope Boundary n — a statement of what lies outside what this dissertation claims
SimCLR	Simple framework for Contrastive Learning of Representations
SIR-n	Source Interpretation Rule n — a rule constraining what may be attributed to a cited source
SSIM	Structural Similarity Index
SSL	Self-Supervised Learning
STARE	Structured Analysis of the Retina (dataset)
UML	Unified Modeling Language
VRAM	Video Random-Access Memory
VVI	Vessel Visibility Index
WSL	Windows Subsystem for Linux
XAI	Explainable Artificial Intelligence
D	Field-of-view diameter, in pixels
CL	Clip limit of the CLAHE transform
F1	F1-score (harmonic mean of precision and recall)
G	Generalization ratio, G = F1_external / F1_EyePACS
T	Global threshold of the dual-constraint CLAHE (T/80 formulation)
α	Class-weight parameter of the Focal Loss (inverse-frequency)
γ	Focusing parameter of the Focal Loss ($\gamma = 2$)
κ	Cohen's Kappa with quadratic weights
σ	Standard deviation of the Gaussian kernel in flat-field correction, $\sigma = 0.07 \cdot D$
ΔF1	Cross-dataset F1 drop, ΔF1 = F1_EyePACS − F1_external